

BATH ZERO

Drive Cycle Generator - Gen 1.5

Point-Mass Simulation

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Abstract

This report details the development and execution of a high-fidelity lap time simulation designed for Bath Zero's electric racing motorcycle. Utilising a comprehensive 3D track model that accounts for localized elevation, pitch, and cross-sectional banking, the solver establishes a Minimum Curvature Path (MCP) to define the vehicle's geometric boundaries. The core physics engine employs a dynamic point-mass vehicle model featuring a non-linear tyre friction algorithm, which realistically scales the contact patch and grip capacity with the motorcycle's instantaneous lean angle. To navigate the circuit at the absolute physical limit, a backwards-pass Velocity Limit (VLIM) profile bounds a forward-looking bang-bang driver model. This longitudinal force application is evaluated using a predictor-corrector numerical integration approach (Heun's) to guarantee solver stability against velocity-dependent aerodynamic drag. The results demonstrate highly accurate longitudinal force constraints, realistic kinematic limit adherence, and precise tracking of mechanical and electrical energy consumption. This simulation provides a robust, data-driven foundation for optimising powertrain ratios, sizing battery capacities, and generating synthesised driver telemetry for vehicle setup tuning.

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1 Introduction

The Gen 1.5 Drive Cycle Generator is a high-fidelity simulation tool designed to bridge the gap between a point-mass model (Gen 1) and full vehicle dynamics (Gen 2). While previous iterations relied on flattened 2D track representations, Gen 1.5 utilises the full 3D track.

The simulation environment is built upon a two-stage computational pipeline. The process begins with the geometry solver, which transforms 5 data points taken along a set separation from a high fidelity map for Assetto Corsa (adapted from LiDAR of the track) into a continuous, optimised trajectory. This trajectory is then used by the physics engine, which treats the motorbike as a dynamic system subject to non-linear constraints such as aero-thermal limits, variable tire contact patches, and powertrain efficiency maps.

By documenting the underlying methodology, this report serves two purposes:

- **Design Validation:** Providing a "digital twin" of the Aragon circuit to test gear ratios and battery discharge cycles before the physical prototype is designed.
- **Algorithmic Foundation:** Establishing the mathematical framework and input data required for the future transition to Full Vehicle Dynamics (FVD) in Gen 2.

The following chapters detail the two core modules: `processTrack.m`, which handles the track construction, and `Sim.m`, which executes the physics-based lap integration and provides telemetry data.

2 Track Definition & 3D Geometry (`processTrack.m`)

The foundation of the Gen 1.5 simulation is the accurate reconstruction of the MotorLand Aragón circuit. Handled by the `processTrack.m` module, this process converts discrete surface data into a continuous, drivable 3D mathematical model.

2.1 Track Surface Data and Interpolation

The track geometry is sourced from a high-fidelity map adapted from LiDAR scans. To accurately capture the circuit's camber and crown across its width, the spatial data is discretized into five lateral lines running parallel along the track:

- Outer Left
- Inner Left
- Center

- Inner Right
- Outer Right

To ensure seamless lap transitions, the script enforces geometric loop closure. Because raw data points can vary in spacing, the cumulative center-line distance is computed, and all five geometric lines (X , Y , and Z coordinates) are interpolated onto a grid of 1500 segments. This is achieved using PCHIP, which prevents overshoots in the track surface curvature.

2.2 Surface Fitting & Track Banking

Motorcycle cornering dynamics and traction limits are highly sensitive to transverse track camber. Instead of simply connecting the leftmost and rightmost bounds, the script calculates a best fit plane through all five transverse data points at every integration step.

A linear polynomial evaluates the relative lateral distances, which are normalised from 0 to 1 based on the total track width at that specific cross-section. It combines this with the relative heights of the five points to compute a unified slope. The local track banking angle, θ_{bank} , is derived from the total fitted change in elevation (Δz) across the calculated track width (w):

$$\theta_{bank} = \arctan\left(\frac{\Delta z}{w}\right) \quad (1)$$

This guarantees that the motorbike correctly experiences gravitational assistance or penalty when navigating cambered or off-camber turns. Furthermore, the longitudinal elevation profile (Z) is calculated and used in `Sim.m`, by differentiating it with respect to the track distance to compute vertical curvature (K_{vert}), dictating the dynamic normal load changes experienced during crests and compressions.

2.3 3D Minimum Curvature Path (MCP) Optimisation

The optimal racing line is established via the MCP Optimisation [2]. Minimising path curvature allows the vehicle to maximise cornering radii, thereby sustaining higher velocities without exceeding the tyres' lateral friction limits.

The path generation is formulated as a Quadratic Programming (QP) optimisation problem. The algorithm seeks to minimise the discrete second derivative of the spatial path, which acts as a derivative for the total path curvature. Let P_i represent the 3D spatial coordinate at node i , defined by its relative position which is normalised across the track width $\alpha_i \in [0, 1]$:

$$P_i = \begin{bmatrix} x_{in,i} + \alpha_i \Delta x_i \\ y_{in,i} + \alpha_i \Delta y_i \\ z_{in,i} + \alpha_i \Delta z_i \end{bmatrix} \quad (2)$$

where x_{in} and y_{in} represent the geometric coordinates of the rightmost track edge, z_{in} and Δz are the intercept and slope derived from the plane created via a transverse linear regression fit.

The objective function $J(\alpha)$ minimises the sum of the squared magnitudes of the directional variations between consecutive path segments across all N nodes [2]:

$$\min_{\alpha} J(\alpha) = \sum_{i=2}^{N-1} \|P_{i+1} - 2P_i + P_{i-1}\|^2 \quad (3)$$

Expanding this squared L_2 norm isolates the terms with respect to α . The quadratic coefficients create the Hessian matrix (H), while the linear cross-terms representing the track

boundaries create the linear coefficient vector (B). The QP solver is bounded strictly by the track limits ($0 \leq \alpha \leq 1$).

Once the spatially optimised nodes are solved, the local cornering radius R is geometrically determined using the circumradius formula for the triangle formed by three consecutive trajectory points:

$$R = \frac{abc}{4A} \quad (4)$$

where a , b , and c are the Euclidean distances between the three adjacent points, and A is the area of the triangle.

To ensure parity with the real-world circuit, all the spatial coordinates and corner radii are scaled to align the simulated lap length exactly with MotorLand Aragón's official distance of 5078 meters. The resulting array of 3D coordinates, radii, and banking angles forms the definitive drive cycle passed to the physics engine.

3 Parameters & Velocity Limit Solver (1.X-2.X Sim.m)

The physical parameters of the motorcycle and its interaction with the track are defined prior to the main dynamic simulation. To ensure the bike navigates the circuit without exceeding physical grip or geometric limits, a pre-calculated Velocity Limit (VLIM) profile is generated. This acts as the boundary condition for the forward-time integration solver.

3.1 Vehicle Specifications and Dynamic Parameters

The motorcycle is modelled as a lumped-mass point vehicle with specialised kinematic behaviours unique to two-wheeled dynamics. The static mass (m) is scaled by an effective mass factor ($M_{eff} = 1.1m$) to account for the rotational inertia of the wheels and powertrain components.

Aerodynamic drag is evaluated using a standard quadratic drag model driven by the frontal area (A) and drag coefficient (C_d). Furthermore, the model implements a dynamic tyre contact patch area derived from the scarce specifications provided by Bridgestone which are reverse engineered using assumptions of 200,000 pa pressure for MotoStudent tyres [5] and the combined mass of the bike and rider being 220kg. The contact area was calculated to scale non-linearly with the motorcycle's lean angle [3, 4] (ϕ). The effective tyre friction coefficient (μ_{eff}) is adjusted proportionally to this contact patch size, realistically modelling the edge-grip characteristics of the racing slicks.

To run the simulation, the following empirical constants and geometric limits were established:

Table 1: Vehicle Mass, Aerodynamics, and Environment

Parameter	Value	Unit
Static Mass (Bike + Rider)	220	kg
Effective Mass Scaling	1.1	-
Center of Gravity Height (h_{cog})	0.45	m
Frontal Area (A)	0.30	m ²
Drag Coefficient (C_d)	0.40	-
Air Density (ρ)	1.225	kg/m ³
Rolling Resistance (A_d, B_d)	0.004, 0.000025	-

Table 2: Tyre Dynamics and Kinematic Limits

Parameter	Value	Unit
Base Friction Coefficient (μ_{base})	1.4	-
Tyre Thickness (t_{tyre})	67.8	mm
Wheel Radius (R_{wheel})	0.3005	m
Maximum Lean Angle	55	°
Maximum Lean Rate	30	°/s

Table 3: Powertrain and Braking System Specifications

Parameter	Value	Unit
Final Drive Ratio	3.68	-
Mechanical Efficiency (Direct Drive)	96	%
Mechanical Efficiency (Geared)	93	%
Auxiliary Power Draw (P_{aux})	150	W
Rotor Outer / Inner Diameter	320 / 246	mm
Caliper Pistons	4 × 33.9	mm
Maximum Line Pressure (P_{line})	1.0	MPa
Pad Friction Coefficient (μ_{pad})	0.4	-

Table 4: Motor Performance Map (Condensed Lookup Table)

Motor Speed (RPM)	Max Torque (Nm)	Efficiency (η_{max})
0	120.00	0.0000
1000	120.00	0.9166
2000	120.00	0.9246
3000	120.00	0.9290
4000	120.00	0.9299
5000	91.67	0.9272
6000	76.39	0.9210
7000	65.48	0.9112
7500	61.12	0.9050

Table 5: Tyre Contact Area vs. Lean Angle

Lean Angle (°)	Contact Area (m ²)
0	0.0054
5	0.0054
10	0.0054
15	0.0055
20	0.0055
25	0.0056
30	0.0057
35	0.0058
40	0.0059
45	0.0061
50	0.0063
55	0.0065

The mechanical braking limit (F_{brake}) applied at the front wheel is derived from the hydraulic system parameters:

$$F_{brake} = \frac{2\mu_{pad}P_{line}A_{piston}R_{eff}}{R_{wheel}}$$

where μ_{pad} is the brake pad friction coefficient, P_{line} is the maximum hydraulic line pressure, A_{piston} is the total caliper piston area, and R_{eff} is the effective rotor radius.

3.2 VLIM: Geometric & Friction Forward Pass 2.1

The first stage of the VLIM generation iterates forward through the track to determine the absolute maximum cornering speed at each node [6]. This steady-state limit is constrained by two primary factors: tyre friction and the physical geometry of the motorcycle.

For any given corner radius (R) and local track banking angle (θ_{bank}), the maximum velocity permitted by the tyre's lateral friction limit (μ_{dyn}) is derived from the banked cornering force balance:

$$v_{friction} = \sqrt{gR \left(\frac{\mu_{dyn} + \tan \theta_{bank}}{1 - \mu_{dyn} \tan \theta_{bank}} \right)}$$

Simultaneously, the speed is bounded by the motorcycle's maximum geometric lean angle (ϕ_{max}). The solver calculates the effective lean relative to gravity, it combines the bike's chassis lean and the track's banking to establish the geometric speed limit:

$$v_{lean} = \sqrt{gR \tan(\phi_{max} + \theta_{bank})}$$

At each node, the maximum allowable velocity is securely clamped to the most restrictive of these two limits, or the vehicle's absolute top speed constraint.

3.3 VLIM: Kinematic Backward Pass & Traction Circle 2.2

While the forward pass dictates how fast a corner can be taken, it does not account for the braking distance required to achieve that speed from a straight. To resolve this, a three-iteration backward pass calculates the deceleration profile recursively from the end of the track to the start [6].

The iterative approach is required as it resolves the boundary condition at the start/finish line, ensuring that the velocity at the final node N is kinematically consistent with the entry velocity at node 1. This prevents artificial speed spikes at the lap transition.

At each integration step moving backward, the solver computes the available longitudinal tyre grip using a standard Traction Circle model [3, 7]. The dynamic normal load (F_z) is comprehensively calculated to include gravitational components, centripetal loading from banking, and vertical aerodynamic/geometric forces driven by the track's vertical curvature (K_{vert}):

$$F_z = mg \cos \theta_{bank} \cos \theta_{pitch} + m \frac{v^2}{R} \sin \theta_{bank} + mv^2 K_{vert}$$

The lateral acceleration required to hold the trajectory consumes a portion of the tyre's total force capacity. The remaining grip dictates the maximum longitudinal deceleration (a_{grip_max}):

$$a_{grip_available} = \sqrt{\left(\frac{\mu_{eff} F_z}{M_{eff}} \right)^2 - a_{lat}^2}$$

The actual deceleration capability is then clamped by the mechanical braking system limit. The total deceleration (a_{total}) aggregates this braking force with resistive forces including aerodynamic drag, rolling resistance, and gravitational track inclines. The allowable entry velocity (v_i) for the preceding node is integrated kinematically over the segment distance (Δs):

$$v_i = \sqrt{v_{i+1}^2 + 2a_{total}\Delta s}$$

This ensures the final VLIM profile represents a physically drivable limit that respects traction thresholds during both cornering and heavy braking zones.

4 Physics Simulation & Force Integration (3.X Sim.m)

With the physical limits established by the VLIM profile, the core simulation executes a forward-time, spatial integration. This solver iterates through the minimum curvature path, computing dynamic state variables and powertrain demands using a predictor-corrector numerical method.

4.1 Kinematic Lean Dynamics 3.2-3.3

The instantaneous lean angle (ϕ) of the motorcycle is highly dynamic and dictates the bike's stability. The foundational roll requirement on a flat surface is derived from the balance of centripetal and gravitational forces [7]:

$$\phi_{flat} = \arctan\left(\frac{v^2}{gR}\right)$$

This ideal lean is immediately corrected for the track's local banking angle (θ_{bank}), which geometrically assists or penalises the required roll. Furthermore, as we utilise thick-profile slick tyres, the shift in the contact patch away from the centreline requires an additional lean angle to maintain the centre of gravity (h_{cog}) at the correct equilibrium. This offset ($\Delta\phi$) is approximated as:

$$\Delta\phi = \arcsin\left(\frac{t_{tyre} \sin(\phi_{ideal})}{h_{cog} - t_{tyre}}\right)$$

To prevent non-physical instantaneous snapping, the rate of change of the roll angle is clamped by a maximum lean rate ($\dot{\phi}_{max}$). Additionally, a tapering function smoothly attenuates this allowed roll rate as the vehicle approaches its maximum physical lean limit, mimicking the smooth inputs of a real rider stabilizing the chassis mid-corner.

4.2 Tyre Dynamics & Traction Limits 3.4-3.5

The simulated tyre model recognizes that the friction coefficient is not static [4, 3]; it relies heavily on the tyre's contact patch area ($A_{contact}$), which expands as the motorcycle leans onto the broader shoulder of the slick tyre. The effective friction coefficient (μ_{eff}) is non-linearly scaled against a reference upright area (A_0):

$$\mu_{eff} = \mu_{base} \left(\frac{A_{contact}}{A_0}\right)^{0.15}$$

Simultaneously, the 3D normal force (F_z) is computed, capturing static weight, track pitch, track banking, and the dynamic vertical load induced by the track's vertical curvature (K_{vert}):

$$F_z = mg \cos \theta_{bank} \cos \theta_{pitch} + m \frac{v^2}{R} \sin \theta_{bank} + mv^2 K_{vert}$$

The total force capacity of the tyre is simply $F_{tire} = \mu_{eff} F_z$. However, cornering consumes a portion of this capacity laterally (F_{lat_demand}). Using a traction circle model, the remaining longitudinal force capacity (F_{long_cap}) available for acceleration or braking is isolated via the Pythagorean theorem:

$$F_{long_cap} = \sqrt{F_{tire}^2 - F_{lat_demand}^2}$$

4.3 Powertrain & Driver Logic 3.6-3.7

The simulated driver operates as a forward-looking bang-bang controller, aggressively targeting the velocity limits defined by the VLIM profile [6]. To reach the target speed of the next spatial node (v_{target}), the required kinematic acceleration (a_{req}) is computed:

$$a_{req} = \frac{v_{target}^2 - v^2}{2\Delta s}$$

The total commanded force (F_{cmd}) required at the contact patch to achieve this acceleration includes the inertial force and all resisting environmental forces:

$$F_{cmd} = a_{req} M_{eff} + F_{drag} + F_{roll} + F_{grav}$$

Before this force is applied, it is bounded by physical constraints. If decelerating, the force is bounded by the mechanical braking system's maximum capacity. While a true physical model would also constrain deceleration by F_{long_cap} , enforcing this dynamic grip limit under heavy braking over the simulation's discrete spatial integration steps introduces numerical instability which propagates an initial error of 0.1m/s in the VLIM braking profile into a complete crash/non-compliance despite the motorcycle being perfectly capable of executing the manoeuvre. If accelerating, the actual applied force is clamped by both the tyre's longitudinal grip limit (F_{long_cap}) and the powertrain's mechanical limit (F_{power_limit}).

The derivation of F_{power_limit} adapts to one of two simulated transmission types:

- **Direct Drive:** The motor RPM is kinematically locked to the wheel speed via a fixed chain drive ratio. The instantaneous available torque is interpolated from the provided motor map. A linear torque-tapering function is applied beyond the 7500 RPM redline to prevent numerical chatter and chattering-induced instability within the solver. This approach approximates the soft limit behaviour of a physical motor controller while ensuring a continuous derivative for the integration of the longitudinal force.
- **Multi-Gear Auto-Shifter:** The solver dynamically selects the optimal gear ratio. Upshifts are triggered when the motor speed exceeds 7400 RPM, ensuring peak power utilisation, while downshifts evaluate if the subsequent lower gear would safely drop the motor below 7100 RPM. To accurately reflect real-world mechanical losses, every gear change initiates a shift delay timer (e.g., 50 ms) where the tractive force is completely cut to zero ($F_{power_limit} = 0$). A subsequent cooldown timer enforces a 1.0-second lockout to prevent erratic shifting oscillation between nodes.

Once the available torque is determined and scaled through the active final drive ratio and estimated mechanical efficiency (η_{mech}), the simulation commits the lowest restricting force to the integration step, perfectly mirroring the constraints of a physical drivetrain.

4.4 Numerical Integration (Heun's Method) 3.8-3.9

Because aerodynamic drag scales with the square of the velocity (v^2), the net force acting on the motorcycle changes continuously across the length of a single spatial segment (Δs). Using

a standard explicit Euler integration (calculating drag only at the start of the segment) causes numerical instability and overestimation of terminal speeds [1].

To resolve this, the solver employs Heun's Method:

1. **Predictor Step:** The net force is calculated using the initial velocity (v_{start}) to predict an estimated end velocity (v_{est}).
2. **Corrector Step:** The aerodynamic drag is recalculated at this predicted v_{est} . The solver then averages the drag force from the start and end of the segment to find a highly accurate average net force ($F_{net.avg}$).

The final velocity for the step is integrated using this corrected average acceleration:

$$v_{end} = \sqrt{v_{start}^2 + 2 \left(\frac{F_{net.avg}}{M_{eff}} \right) \Delta s}$$

The time step (Δt) elapsed during the segment is recorded as $\Delta t = \Delta s / v_{mean}$, accumulating into the final lap time. The mechanical power requirement is logged and divided by the powertrain efficiency map, adding the auxiliary electrical load to track the total energy consumed over the lap.

5 Plots & Telemetry (Sim.m 4.X-5.X)

The data output by the sim can be split into two parts. Debug plots and data to ensure the mathematical validity of the model and the regular outputs which can be used for feasibility studies.

5.1 Lap Time & Energy

The total lap time is simply the summation of the individual time steps (Δt) elapsed across all spatial segments. From this, the average vehicle speed (v_{avg}) is derived using the total track length (S_{total}):

$$v_{avg} = \frac{S_{total}}{\sum \Delta t}$$

The energy cost of the lap is also calculated. The simulation tracks the instantaneous mechanical power delivered to the rear wheel (P_{mech}), scales it by the dynamic powertrain efficiency map (η_{active}), and adds the constant auxiliary electrical draw (P_{aux}). The total electrical energy consumed (E_{total}) in Joules is integrated over the lap:

$$E_{total} = \sum \left(\frac{P_{mech}}{\eta_{active}} + P_{aux} \right) \Delta t$$

This value is subsequently converted into Watt-hours (Wh) and kilowatt-hours (kWh) to evaluate battery capacity requirements and calculate the vehicle's specific energy economy (Wh/km).

5.2 Data Visualization & Kinematic Analysis

To validate the model, the solver generates a comprehensive suite of diagnostic plots mapping both the spatial and time domains:

- **Track Geometry & Velocity Mapping:** The trajectory is mapped in 3D space to demonstrate the elevation profiles, while a 2D plot overlays the vehicle's instantaneous velocity gradient. This demonstrates clearly the braking zones and corner apex speeds with the physical track layout.

- **Velocity Traces (VLIM vs. Actual):** A time plot compares the target limit velocity profile against the actual integrated speed. This validates the bang-bang controller and confirms that the numerical integration accurately tracks, and doesn't exceed, the physical grip and geometric boundaries.
- **Force Constraint Verification:** The solver plots the raw commanded longitudinal force alongside the dynamic tyre grip limit (F_{long_cap}) and powertrain capability (F_{power_limit}). This demonstrates where the vehicle is grip-limited versus power-limited during acceleration/braking phases.
- **Powertrain Power Flow:** The mechanical power output of the motor is plotted against the total electrical power drawn from the battery. The delta between these curves visually represents the thermal losses of the system.

5.3 Driver Telemetry

Beyond visualisation, the simulation constructs a standard telemetry array for external analysis. An approximate steering angle (δ) is calculated using the track's spatial curvature (κ), the vehicle's lean angle (ϕ), and the wheelbase (L):

$$\delta = (L \cdot |\kappa|) \cos(|\phi|)$$

This steering trace is combined with normalised throttle (0 to 1) and brake inputs, calculated by taking the ratio of applied longitudinal force to the maximum available capabilities. These arrays are exported as a time-series CSV file (e.g., *Aragon_Telemetry.csv*), providing a direct data bridge to Simulink for Gen 2 Development.

6 Results & Analysis

The following analysis compares the updated high-fidelity model against previous iterations to highlight improvements in dynamic accuracy.

6.1 New Track & Elevation

Understanding vehicle performance requires correlating dynamic states with the physical track geometry. Figure 1 maps the updated minimum curvature trajectory in 2D space. The algorithm successfully identifies the optimal racing line, utilizing the full width of the track to maximize corner radii and increase the geometric velocity limits.

Crucially, the track is not treated as a 2D plane. Figure 2 illustrates the pronounced elevation changes and steep gradients present on the circuit. This steepness explicitly justifies the necessity of the 3-axis normal force (F_z) calculation in the physics solver. By accounting for local pitch, banking angles, and the dynamic vertical load induced by vertical curvature (K_{vert}), the model prevents severe overestimations of tyre grip that would occur in a simplified planar model.

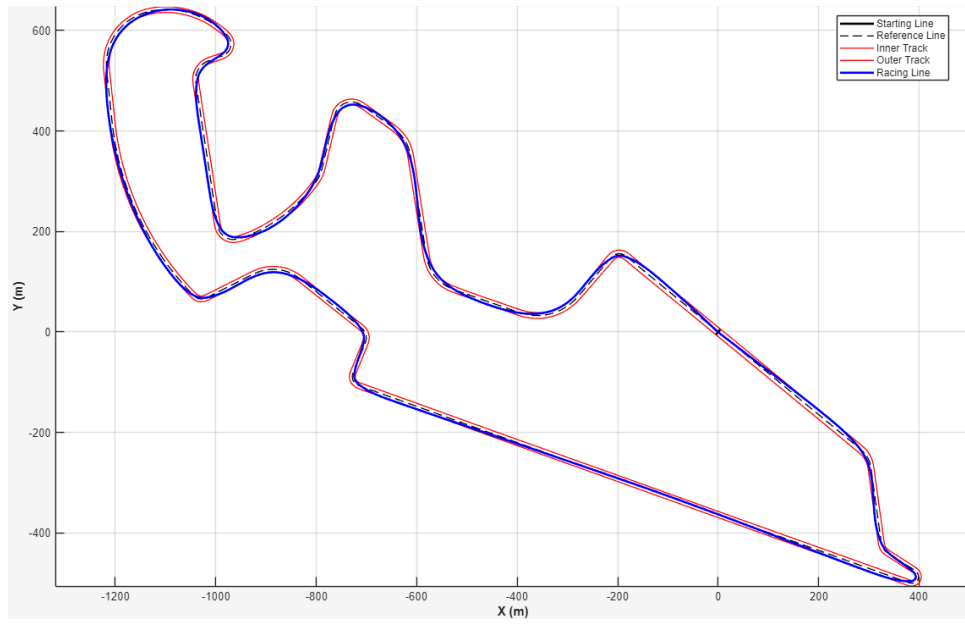


Figure 1: Updated track boundaries and the computed Minimum Curvature Path (MCP).

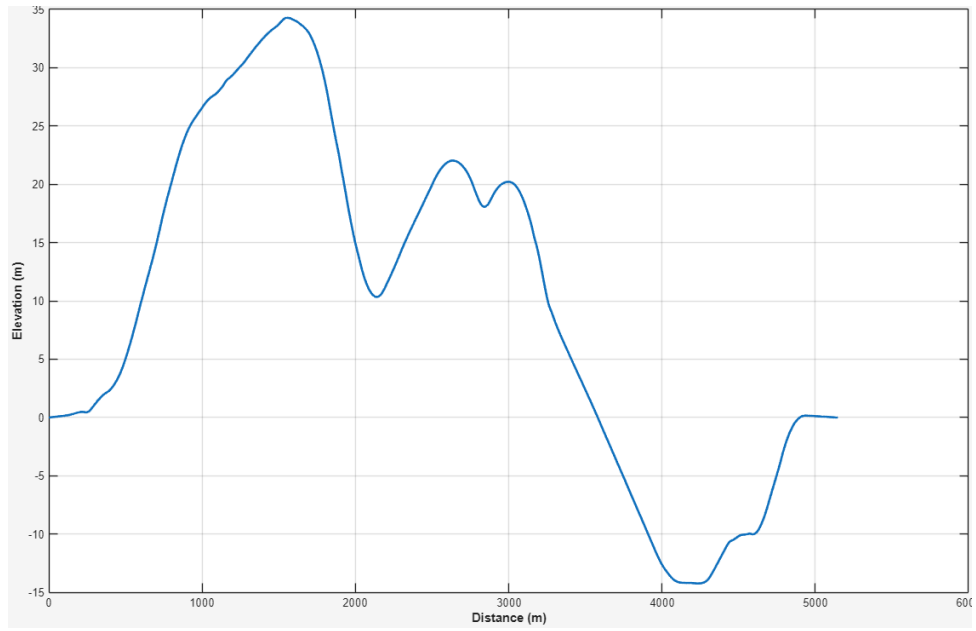


Figure 2: 3D elevation profile of the circuit.

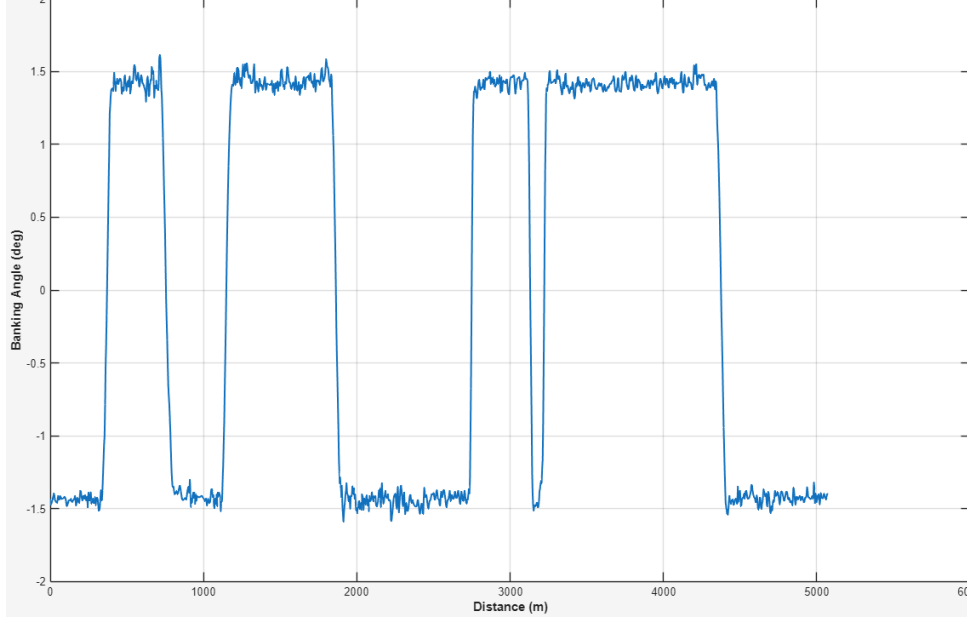


Figure 3: Cross-sectional banking angle variation along the circuit.

Before evaluating the dynamics, it is critical to demonstrate the necessity of a high-definition track for current and future models. Figure 3 demonstrates that the circuit’s cross-sectional camber continuously fluctuates between 1.5° and -1.5° . Because motorcycle cornering limits are highly sensitive to local banking, where positive camber assists the required roll angle and negative camber penalizes it, relying on a single 1D centreline extraction via GPS is insufficient and will lead to incorrect conclusions.

These results explicitly justify the model’s approach of scanning five distinct points across the track width. By capturing the localized banking variations away from the centreline, the solver ensures that the dynamic friction and normal force calculations remain highly accurate regardless of where the computed Minimum Curvature Path (MCP) positions the vehicle laterally on the track.

6.2 Velocity Gradient Comparison

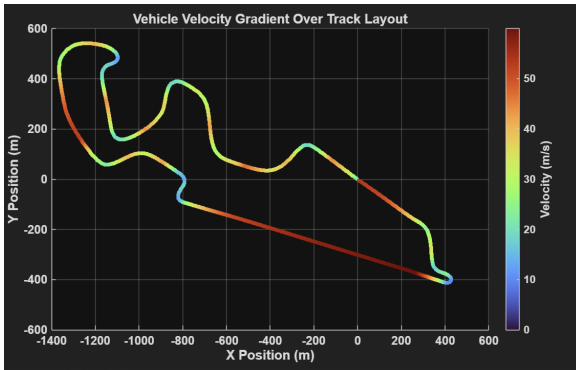


Figure 4: Previous spatial velocity mapping.

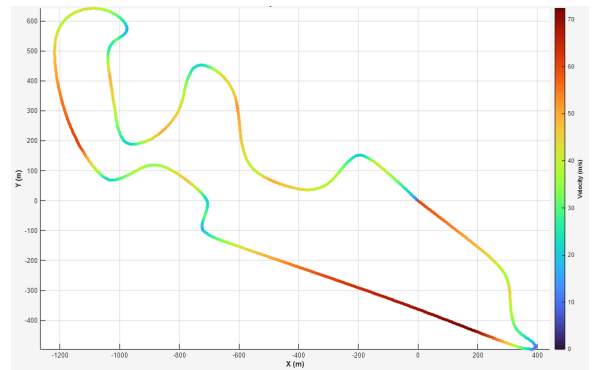


Figure 5: Updated spatial velocity mapping.

Figures 4 and 5 overlay the vehicle’s instantaneous velocity gradient directly onto the track map. Comparing the updated model against the previous iteration reveals a much smoother, more realistic distribution of speed. The updated solver correctly carries higher apex speeds

due to the dynamic tyre friction model and initiates braking phases more accurately in response to the track's 3D geometry.

6.3 VLIM Tracking

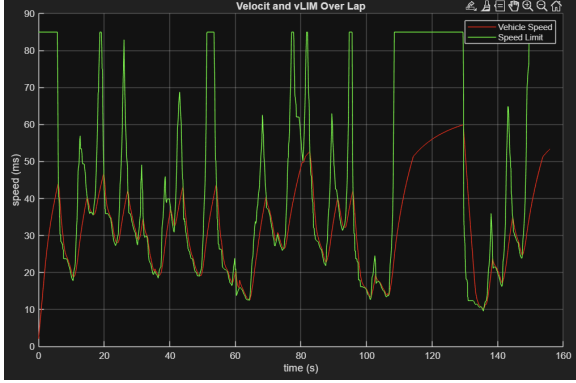


Figure 6: Previous VLIM tracking profile.

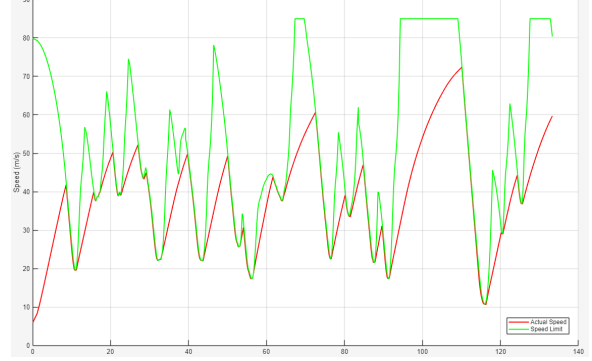


Figure 7: Updated VLIM tracking profile.

A critical measure of the simulation's robustness is its ability to aggressively target, but never exceed, the physical limits of the motorcycle. Figures 6 and 7 plot the integrated vehicle speed against the theoretical Velocity Limit (VLIM) profile.

In the updated model (Figure 7), the upgrades to the tyre models and the new backwards pass is evident. The VLIM is smoother, less jagged and allows the motorcycle to carry more speed than before. The velocity trace also demonstrates smooth, precise deceleration curves that perfectly ride the VLIM boundary entering corners, compared to the overshoots present in the previous iteration. The vehicle safely maximises acceleration on corner exit without breaching the geometric or friction limits.

6.4 Longitudinal Force Bounding

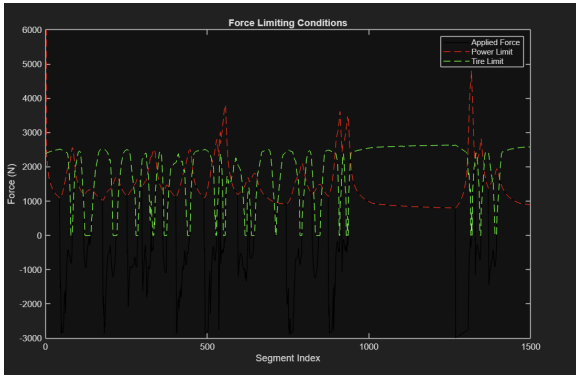


Figure 8: Previous longitudinal force constraints.

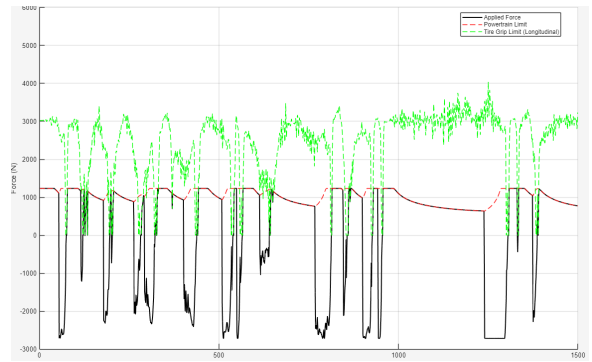


Figure 9: Updated longitudinal force constraints.

To confirm the realism of the longitudinal dynamics, Figures 8 and 9 overlay the applied force against the powertrain limit (F_{power_limit}) and the tyre's remaining longitudinal grip (F_{long_cap}).

Although the overall graph remains similar, the new model exhibits increased oscillated and chattering. Which can be attributed to the high resolution of the track data and tyre friction mapping.

6.5 Synthesized Driver Telemetry

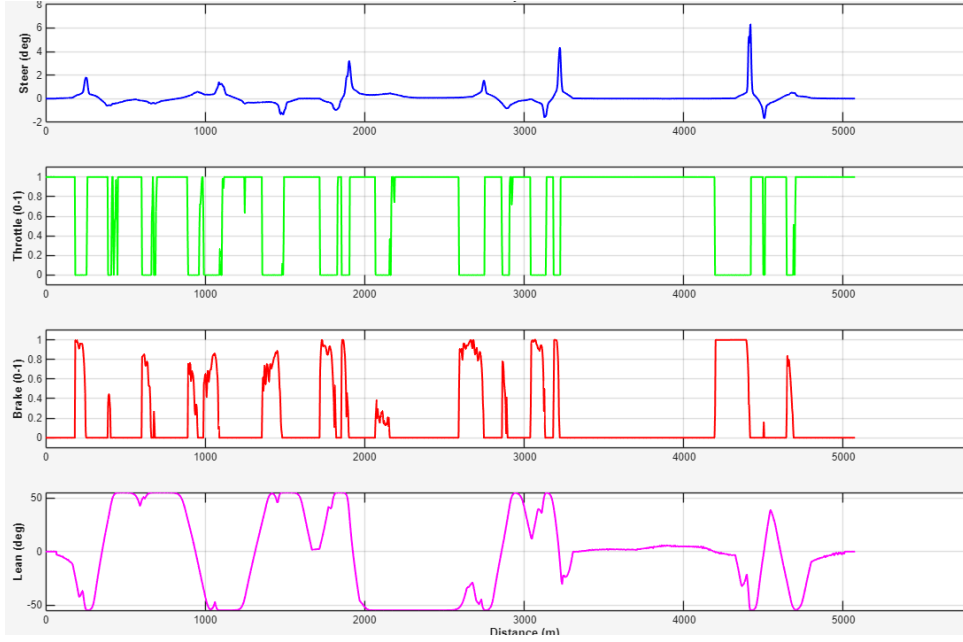


Figure 10: Driver inputs: Steering, Throttle, Brake, and Lean Angle.

Figure 10 displays the calculated telemetry for the driver inputs. Normalized throttle (0 to 1) and brake inputs accurately reflect the force bounds discussed previously.

Additionally, the trace illustrates the smooth, continuous application of the lean angle (ϕ) and the less smooth, approximated steering angle (δ). This proves the tapering function applied in the physics solver is functioning correctly, preventing non-physical, instantaneous snapping from left to right. This telemetry output allows the simulated lap to be loaded into simulink for the development of Gen 2.

7 Model Limitations & Future Work

While the current numerical solver establishes a highly stable and mathematically rigorous physical baseline, it predicts a theoretical best-case scenario. To bridge the gap between this absolute physical limit and a human-achievable lap time, several idealisations within the driver model and vehicle parameters could be addressed in future iterations.

7.1 Driver Model

Currently, the simulated rider operates as a forward-looking bang-bang controller. This deterministic approach aggressively commands maximum permissible acceleration or deceleration to ride the extreme boundary of the VLIM profile. While efficient, it lacks human physiological constraints.

The model, using a pre-calculated VLIM is accurate to industry practice and mimics a driver being briefed on the track layout prior to a race. It is an effective system and should be kept. However, future work could transition from a bang-bang system to a proportional-integral-derivative (PID) or pure proportional controller. By tuning the controller gains to mimic human inputs, the force application will become smoother and less absolute. Additionally, incorporating a discrete human reaction time delay (t_{react} , typically 200–300 ms) before initiating heavy

braking phases would significantly increase the realism of corner entry dynamics, resulting in a more conservative and achievable predicted lap time.

7.2 Aerodynamics

The model is highly optimistic regarding aerodynamic resistance. It relies on a static drag coefficient (C_d) and a constant frontal area (A_{front}) regardless of the vehicle’s dynamic state. In reality, a motorcycle’s aerodynamics are heavily transient.

Future iterations could implement a dynamic aerodynamic map. This would scale A_{front} and C_d as a function of both the instantaneous velocity (simulating a rider tucking in at high speeds) and the lean angle (ϕ), as leaning shifts the physical profile exposed to the streamline of air.

7.3 Powertrain & Mechanical

The simulated powertrain operates under highly optimistic assumptions that do not fully penalize transient mechanical states:

- **Motor Efficiency:** The solver interpolates available torque and electrical efficiency (η_{mech}) from ideal, steady-state motor maps. It currently ignores thermal effects. Prolonged high-current draw during a lap would realistically induce thermal derating, reducing the peak available torque (T_{max}) over time. Future models could integrate a basic lumped-mass thermal model to dynamically scale torque limits.
- **Transmission Dynamics:** The multi-gear auto-shifter utilizes a 50 ms shift delay ($F_{power_limit} = 0$). For a mechanical, non-seamless gearbox, this time lag is exceptionally short. Real-world clutch actuation, gear engagement, and throttle blipping typical of human riders dictate a shift lag closer to 150–250 ms. Expanding this delay will compound over the lap, providing a more accurate penalty for multi-gear setups compared to direct-drive configurations.

By addressing these limits, softening the driver inputs and applying realistic penalties to the aerodynamics and powertrain, the simulation will evolve from calculating the absolute physical limit of the motorcycle to accurately predicting the achievable race pace of a human rider. Which would be much more useful later down the line. The current model sits as a perfect foundation for any further modifications to align it into such a system.

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